

Empirical Tools of Public Finance

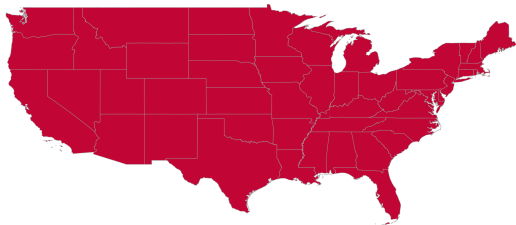
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Do Pollution Regulations Work?

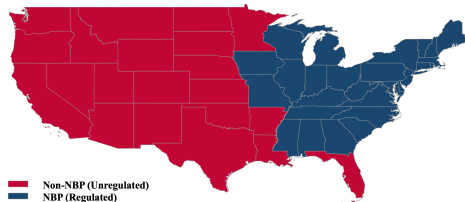
- In Public Economics, we often need to know the effects of a government policy
- Example: 2003 limit to summer air pollution (NO_x) in states like Tennessee



2003 Policy Reform: Regulation in High Polluting States



Winter months (NBP vs Non-NBP)



Summer months (NBP vs Non-NBP)

Source: Deschenes-Greenstone-Shapiro (AER 2017)

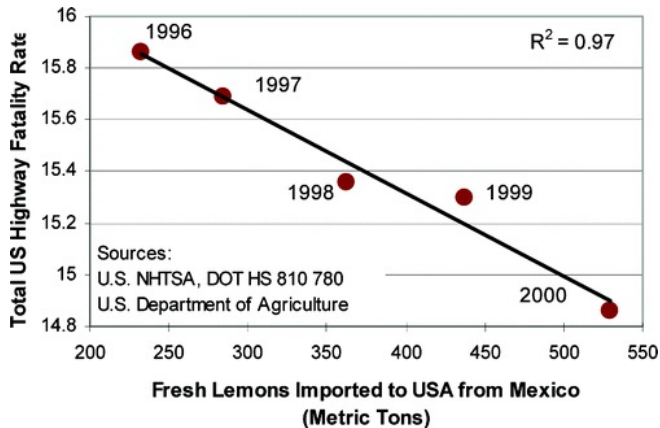
- NBP: NO_x Budget Program (“cap-and-trade”)

- **Empirical public economics:** The use of data and statistical methods to measure the impact of government policy on individuals and markets (example: how the 2003 pollution regulation affected air pollution)
- **Correlation:** Two economic variables are correlated if they move together (example: height and weight across individuals)
- **Causality:** Two economic variables are causally related if the movement of one causes movement of the other (example: good nutrition as an infant increases adult height)

More Specifically: But-For Causality

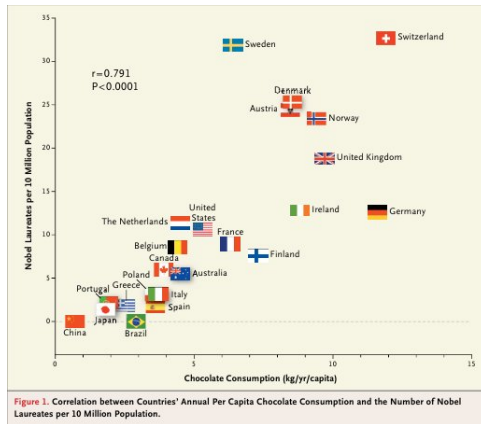
- Causality is not about fault or guilt
- Causality is a “but-for” test. X caused Y if the following statement is true: “But for the action X, the result Y would not have happened.”
- Equivalently (and more naturally), causality means: **“If action X had not happened, then the result Y would not have happened.”**
- A single outcome can have many causes

Have Lemon Imports Reduced Car Accident Deaths?



Source: Johnson (2008) *American Chemical Society*

Does Chocolate Cause Nobel Prizes?



Source: Messerli (2012) *New England Journal of Medicine*

The Identification Problem

- The attempt to interpret a correlation as a causal relationship without sufficient thought to the underlying process generating the data is a common problem
- For any correlation between two variables A and B, there are three possible explanations, one or more of which could result in the correlation:
 - ① A is causing B
 - ② B is causing A
 - ③ Some third factor is causing both
- The general problem that empirical economists face in trying to use existing data to assess the causal influence of one factor on another is that one cannot immediately go from correlation to causation

Randomized Trials as a Solution

- **Randomized trial:** The ideal type of experiment designed to test causality, whereby a group of individuals is randomly divided into a treatment group, which receives the treatment of interest, and a control group, which does not
- **Treatment group:** The set of individuals who are subject to an intervention being studied
- **Control group:** The set of individuals comparable to the treatment group who are not subject to the intervention being studied.
- Randomized trials have been used in medicine for many decades and have become very popular in economics, especially development economics since the turn of the century

The Problems of Bias

- **Bias:** Any source of difference between treatment and control groups that is correlated with the treatment but is not due to the treatment
- Bias from small samples: Having large sample sizes allows researchers to eliminate any consistent differences between groups by relying on the statistical principle called *the law of large numbers*: the odds of getting the wrong answer approaches zero as the sample size grows
- Statisticians develop methods to evaluate the precision of estimates and create confidence intervals around estimates
- But bias often remains in large samples, unless it is a randomized trial

Examples of Randomized Trials

- **Randomized Trials of statins (a drug that lowers cholesterol)**

- 1994 Scandinavian Simvastatin Survival Study: The randomized trial of the Simvastatin drug tracked 4,000 people with heart disease across five countries. Treatment group: 2,000 were randomly selected to get the drug. Control group: the remaining 2,000 got a sugar pill. The treatment group enjoyed a 30% reduction in mortality over the control group after five years.

- **Randomized Trials in the Welfare Reform (TANF) Context**

- Randomized trials are equally useful in the context of public policy. A number of states implemented randomized trials to test various aspects of proposed welfare reform.

Why We Need to Go Beyond Randomized Trials

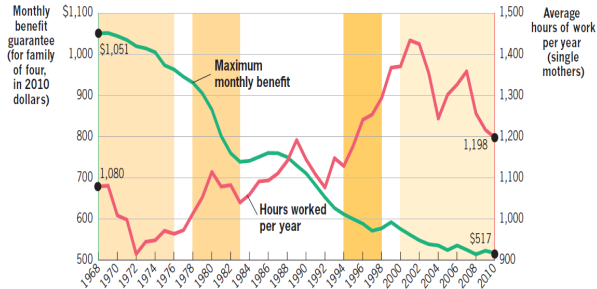
Even the gold standard of randomized trials has some potential problems.

- **1) External validity:** The results are only valid for the sample of individuals who volunteer to be either treatments or controls, and this sample may be different from the population at large (e.g., randomized experiment in Sweden or US would not necessarily generate the same results)
- **2) Attrition:** Individuals may leave the experiment before it is complete. Reduction in the size of samples over time, which, if not random, can lead to bias estimates.
- Outside randomized experiments, bias is a pervasive problem that is not easily remedied. There are, however, methods available that can allow us to approach the gold standard of randomized trials.

- **Observational data:** Data generated by individual behavior observed in the real world, not in the context of deliberately designed experiments.
- **Time series analysis:** Analysis of the co-movement of two series over time.
- **Cross-sectional regression analysis:** Statistical analysis of the relationship between two or more variables exhibited by many individuals at one point in time.

3.3

Time Series Analysis: Cash Welfare Guarantee and Hours Worked Among Single Mothers



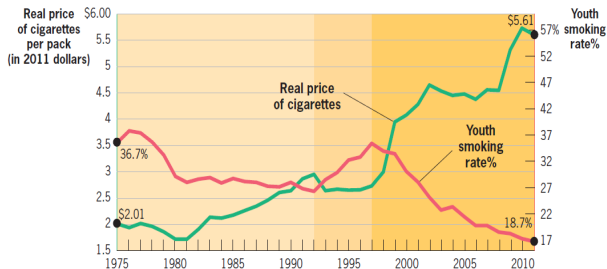
Time series analysis: Analysis of two series over time.

Problems with Time Series Analysis

- Although this time series correlation is striking, it does not necessarily demonstrate a causal effect of TANF benefits on labor supply
- When there is a slow-moving trend in one variable through time, as is true for the general decline in income guarantees over this period, it is very difficult to infer its causal effects on another variable.
- Other factors get in the way of a causal interpretation of this correlation over time; factors such as economic growth and a more generous Earned Income Tax Credit (EITC) can cause bias in this time series analysis because they are also correlated with the outcome of interest.

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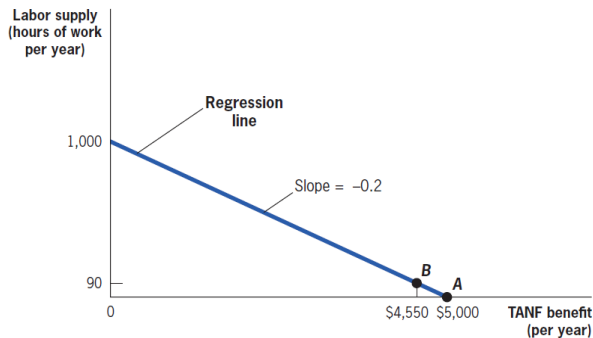
When Is Time Series Analysis Useful? Cigarette Prices and Youth Smoking



- Sharp, simultaneous changes in prices and smoking rates in 1993 and 1998–onward
- Known causes: price war, tobacco settlements

3.3

Cross-Sectional Regression Analysis: Labor Supply and TANF Benefit



Regression line: The line that measures the best linear approximation to the relationship between any two variables.

$$Y = X\beta + \varepsilon$$

- X is the independent variable data (TANF benefit guarantee)
- Y is the dependent variable data (labor supply)
- β is the coefficient that measures the effect of X on Y ε is the error term (captures variations in Y not related to X).
- Ordinary least square regression (OLS) estimates β without bias if ε is not correlated with X

- The estimated coefficient $\hat{\beta}$ is reported with standard errors in parentheses
- Example: $\hat{\beta} = 0.5$ (0.1) should be understood as β is in confidence interval $(0.5 - 2 \times 0.1, .5 + 2 \times 0.1) = (0.3, 0.7)$ with probability 95%.
- We have standard errors because we do not know the exact value of β
- When the estimated coefficient is more than twice the standard error, we can conclude that it is significantly different from zero (i.e., we can reject the hypothesis of no effect with 95% confidence).

Problems with Cross-Sectional Regression Analysis

- The result summarized in Figure 3-4 seems to indicate strongly that mothers who receive the largest TANF benefits work the fewest hours. Once again, however, there are several possible interpretations of this correlation.
- One interpretation is that higher TANF benefits are causing an increase in leisure.
- Another possible interpretation is that in places with high TANF benefits, mothers have a high taste for leisure and wouldn't work much even if TANF benefits weren't available (this means exactly that ε is correlated with X)

Control Variables

- It is essential in all empirical work to ensure that there are no factors that cause consistent differences in behavior across two groups (ε) and are also correlated with the independent variable X
- **Control variables:** Additional variables Z that are included in cross-sectional regression models to account for differences between treatment and control groups that can lead to bias

$$Y = X\beta + Z\gamma + \varepsilon$$

- In TANF case, Z would include race, education, number of children to control for demographic differences across states
- Empirically, add Z variables and assess whether they change the estimate β . If estimate β varies a lot, we cannot be confident that identification assumption holds

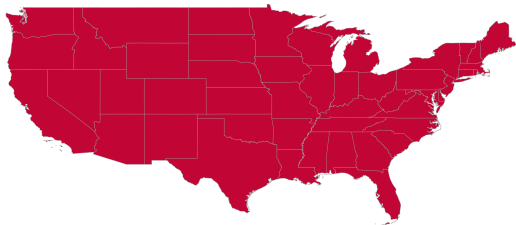
Natural Experiments: Definition

- **Natural experiments** (also called quasi-experiments) Changes in the economic environment that create nearly identical treatment and control groups for studying the effect of that environmental change, allowing public finance economists to take advantage of as-good-as-random assignment created by external forces
- Example: The 2003 NO_x Budget Program (NBP) regulates summer emissions in certain states but not others. NBP states in summer are the Treatment (T) group; non-NBP states (or winter months) serve as the Control (C) group. We'll explore this in detail.

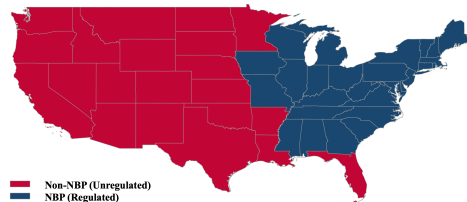
Case Study: Did the 2003 NBP Program Actually Reduce Pollution?

- **Policy / setting:** The NO_x Budget Program (NBP) is a cap-and-trade program that targets NO_x emissions from power plants during the *summer (ozone season)*
- **Question:** Did the NBP *actually reduce pollution*?
- **Why this is a natural experiment:**
 - The policy is summer-focused, so winter months can serve as a potential control group
 - The policy affected only some states, so other states can serve as a potential control group

2003 Policy Reform: Regulation in High Polluting States



Winter months (NBP vs Non-NBP)



Summer months (NBP vs Non-NBP)

Source: Deschenes-Greenstone-Shapiro (AER 2017)

- NBP: NO_x Budget Program (“cap-and-trade”)

Let's Go to the Data!

Natural Experiments: Estimation of the Causal Effect

- We consider a Treatment group (T) and a Control group (C) and outcome Y
- **Simple difference estimator:** $D = Y^{T,After} - Y^{C,After}$ is the difference in average outcomes between treatment and control after the change
- In randomized experiment, simple difference $D = Y^{T,After} - Y^{C,After}$ is sufficient because T and C are identical before the treatment
- In natural experiment, T and C might not be comparable before treatment. You can compute $D^{before} = Y^{T,Before} - Y^{C,Before}$
- If $D^{before} = 0$, you can be fairly confident that $D = Y^{T,After} - Y^{C,After}$ estimates the causal effect

Difference-in-Difference estimator

- If simple difference $D^{before} = Y^{T,Before} - Y^{C,Before}$ is not zero, you can form the **Difference-in-Difference estimator**

$$DD = [Y^{T,After} - Y^{C,After}] - [Y^{T,Before} - Y^{C,Before}]$$

- This measures whether the difference between treatment and control changes after the policy change
- (Equivalent alternative: simple difference estimator for treatment, minus simple difference estimator for control:)

$$DD = [Y^{T,After} - Y^{T,Before}] - [Y^{C,After} - Y^{C,Before}]$$

- DD identifies the causal effect of the treatment if, absent the policy change, the difference between T and C would have stayed the same (this is called the “parallel trends” or “common trends” assumption)

Measuring Pollution: NO_x Emissions Data

- **Outcome:** NO_x emissions in 1000s of tons from power plants (state × season × year)
- **Data source:** EPA continuous emissions monitoring systems (CEMS)
- **Summer (treatment season):** May–September (NBP in force)
- **Winter (comparison season):** January–April, October–December
- **Prediction:** If policy works, summer NO_x in NBP states should drop after program starts, relative to the winter and relative to non-NBP states

DID Estimate of NOx Budget Program in Tennessee

	Summer	Winter	Difference
Pre-treatment (1997-2002)	77	71	7
Post-treatment (2003-2007)	27	59	-32
Difference	-50	-11	-39

Source: Deschenes-Greenstone-Shapiro (AER 2017)

Problems with Natural Experiments

- With natural experimental studies, we can never be completely certain that we have purged all bias from the treatment–control comparison.
- Natural experimental studies present various robustness checks to try to make the argument that they have obtained a causal estimate.
- Examples: find alternative control groups, do a placebo comparing treatment and control DD when no policy change took place, etc.
- Best way to check validity of DD estimator is to plot times series and assess whether a clear break between the two groups happens at the time of the reform

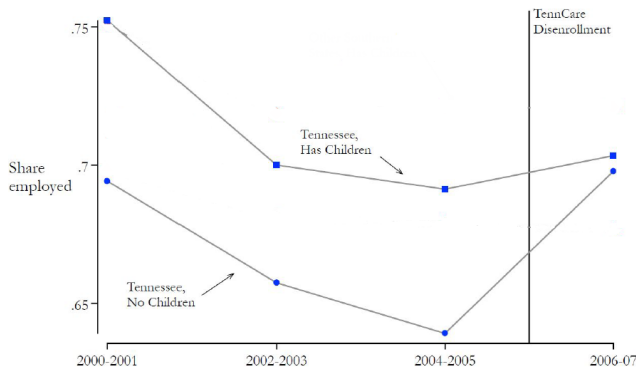
Validating a DD Analysis

- DD “common trends” identifying assumption: In the absence of the treatment, outcomes for the treatment group and control group would have trended similarly.
- Typical check: Do outcomes for the treatment group and the control group trend similarly *before* the treatment?
- But often times, there are only only a few data points from before the treatment. So one does not have tremendous confidence that, in the absence of the treatment, outcomes would have trended similarly.
- So what do economists do when the simple DD is not tremendously compelling?

Validating a DD Analysis

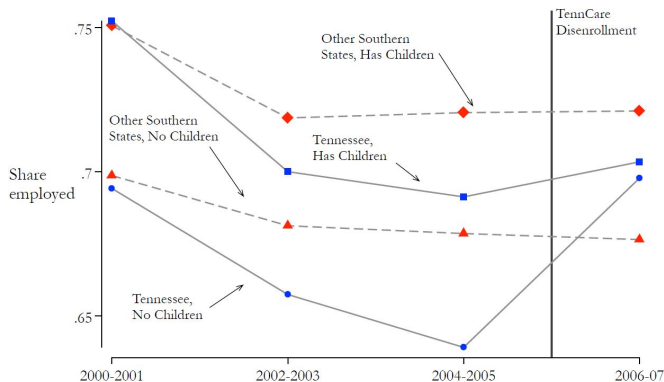
- Additional check called a “placebo test”: Are there similar unaffected groups that trend similarly before and after the treatment?
- Example: Tennessee in 2005 abruptly removed 170,000 adults from Medicaid (free medical care for the poor), especially affecting poor childless adults. Did those childless adults enter the workforce in order to get health insurance from an employer instead?

With Only Annual Data, This DD Could Benefit from a Placebo Test



Source: Garthwaite-Gross-Notowidigdo (QJE 2014)

Placebo Test Using Neighboring States



Source: Garthwaite-Gross-Notowidigdo (QJE 2014)

- The central issue for any policy question is establishing a causal relationship between the policy in question and the outcome of interest.
- We discussed several approaches to distinguish causality from correlation. The gold standard for doing so is the randomized trial, which removes bias through randomly assigning treatment and control groups.
- Unfortunately, however, such trials are not available for every question we wish to address in empirical public economics. As a result, we turn to alternative methods such as time series analysis, cross-sectional regression analysis, and natural experimental analysis.
- Each of these alternatives has weaknesses, but careful consideration of the problem at hand can often lead to a sensible solution to the bias problem that plagues empirical analysis.

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